CNSCC.405, Lab 2

General Path Loss Model

The free-space propagation model is used for predicting the received signal strength in the line-of-sight (LOS) environment where there is no obstacle between the transmitter and receiver. It is often adopted for satellite communication systems. Let d denote the distance in meters between the transmitter and receiver. When non-isotropic antennas are used with a transmit gain of G_t and a receive gain of G_r, the received power at distance d, $P_r(d)$, is expressed by the equation [1.1], given as

接收功率：
$$P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L} \tag{1.1}$$

where P_t represents the transmit power (watts), λ is the wavelength of radiation (m), and L is the system loss factor which is independent of propagation environment. The system loss factor represents overall attenuation or loss in the actual system hardware, including transmission line, filter, and antennas. In general, $L > 1$, but $L = 1$ if we assume that there is no loss in the system hardware. It is obvious from Equation (1.1) that the received power attenuates exponentially with the distance d. The free-space path loss, $PL_F(d)$, without any system loss can be directly derived from Equation (1.1) with L=1, as

$$PL_F(d)[dB] = 10\log\left(\frac{P_t}{P_r}\right) = -10\log\left(\frac{G_t G_r \lambda^2}{(4\pi)^2 d^2}\right) \tag{1.2}$$

Without antenna gains (i.e., $G_t = G_r = 1$), Equation (1.2) is reduced to

$$PL_F(d)[dB] = 10\log\left(\frac{P_t}{P_r}\right) = 20\log\left(\frac{4\pi d}{\lambda}\right) \tag{1.3}$$

Figure 1 shows the free-space path loss at the carrier frequency of f_c=1.5 GHz for different antenna gains as the distance varies. In fact, a more generalized form of the path loss model can be constructed by modifying the free-space path loss with the path loss exponent n that varies

Ref: MIMO-OFDM wireless communications with MATLAB. Cho, Y. S., Kim, J., Yang, W. Y., & Kang, C. G. John Wiley & Sons, 2010.

with the environments. This is known as the log-distance path loss model, in which the path loss at distance d is given as

$$PL_{LD}(d)[dB] = PL_F(d_0) + 10n\log\left(\frac{d}{d_0}\right) \tag{1.4}$$

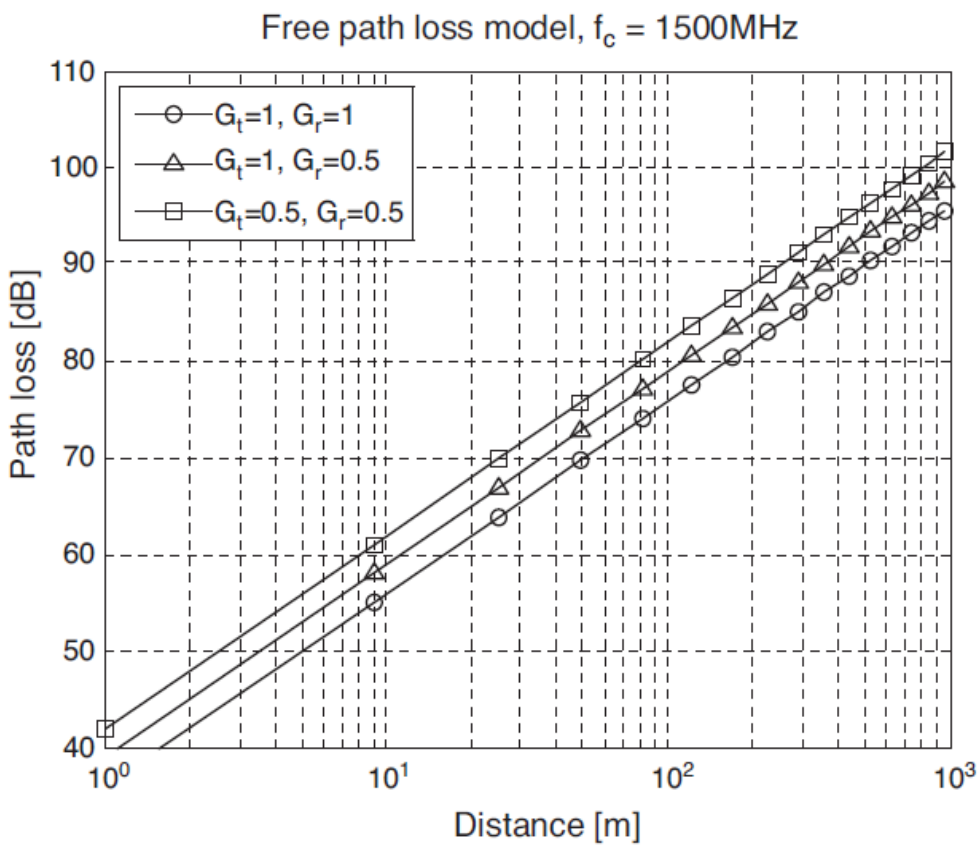

Figure 1: Free-space path loss model.

where d_0 is a reference distance at which or closer to the path loss inherits the characteristics of free-space loss in Equation (1.2). As shown in Table 1.1, the path loss exponent can vary from 2 to 6, depending on the propagation environment. Note that n= 2 corresponds to the free space. Moreover, n tends to increase as there are more obstructions. Meanwhile, the reference distance d_0 must be properly determined for different propagation environments. For example, d_0 is typically set as 1 km for a cellular system with a large coverage (e.g., a cellular system with a cell radius greater than 10 km). However, it could be 100 m or 1 m, respectively, for

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a macro- cellular system with a cell radius of 1km or a microcellular system with an extremely small radius.

Table 1.1 Path loss exponent

Environment	Path loss exponent (n)
Free space	2
Urban area cellular radio	2.7–3.5
Shadowed urban cellular radio	3–5
In building line-of-sight	1.6–1.8
Obstructed in building	4–6
Obstructed in factories	2–3

(Rappaport, Theodore S., *Wireless Communications: Principles and Practice*, 2nd Edition, © 2002, pg. 76. Reprinted by permission of Pearson Education, Inc., Upper Saddle River, New Jersey.)

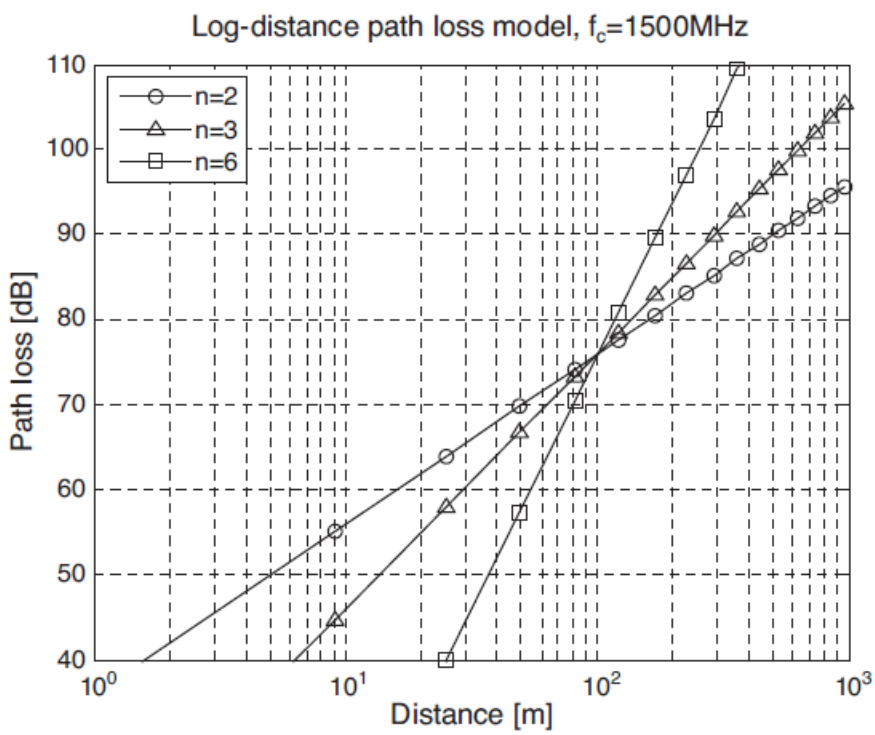

Figure 2: Log-distance path loss model

Figure 2 shows the log-distance path loss by Equation (1.5) at the carrier frequency of $f_c = 1.5$ KHz and $d_0 = 100$m. However, all the aforementioned path loss models do not take this particular situation into account. A *log-normal* shadowing model is useful when dealing with a more realistic situation. Let X_σ denote a Gaussian random variable with a zero mean and a standard deviation of σ. Then, the log-normal shadowing model is given as

Ref: MIMO-OFDM wireless communications with MATLAB. Cho, Y. S., Kim, J., Yang, W. Y., & Kang, C. G. John Wiley & Sons, 2010.

$$PL(d)[dB] = \overline{PL}(d) + X_\sigma = PL_F(d_0) + 10n\log\left(\frac{d}{d_0}\right) + X_\sigma \qquad (1.5)$$

In other words, this particular model allows the receiver at the same distance d to have a different path loss, which varies with the random shadowing effect X_σ. Figure 3 shows the path loss that follows the log-normal shadowing model at f_c = 1.5 GHz with σ =3 dB and n = 2.

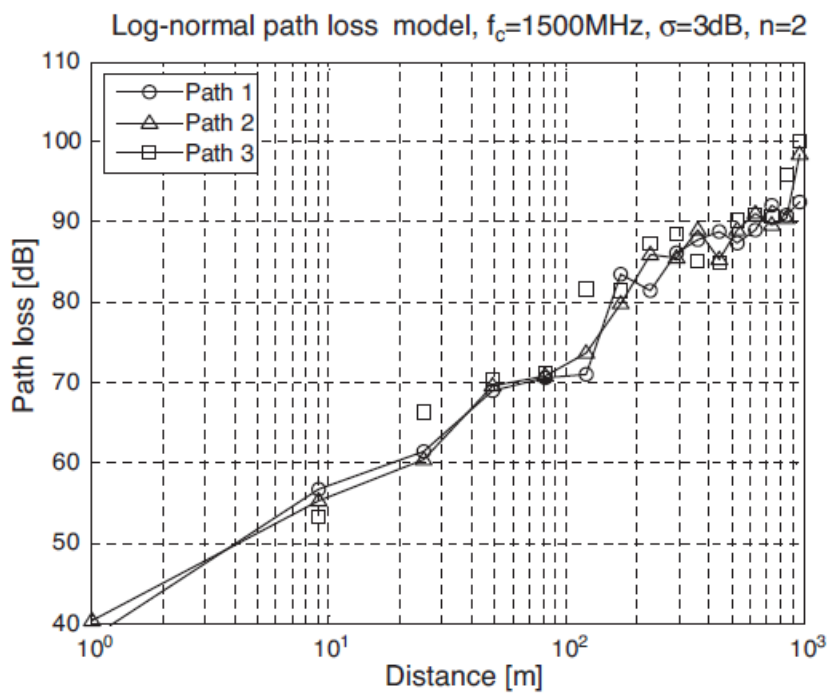

Figure 3: Log-normal shadowing path loss model.

Tasks: 1. 绘制所有考虑模型的路径损耗与距离的函数关系图。（如果需要，请做出任何现实的假设）。根据结果，使用比较分析方法得出您应该考虑的模型的结论。

Plot the path losses as a function of distance for all considered models. (Make any realistic assumptions if required). Draw a conclusion on which model you should consider, based on the results, using comparative analysis method.

Tasks: 2. 写一个总结，说明你学到了什么，观察到了什么，出了什么问题（如果在你的实验中，你没有得到预期的结果），以及你将来如何克服类似的问题。

Write a summary of what you have learned, observed, things went wrong (if in your experiment, you have not had the predicted result), and how you can overcome the similar problem in future.

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